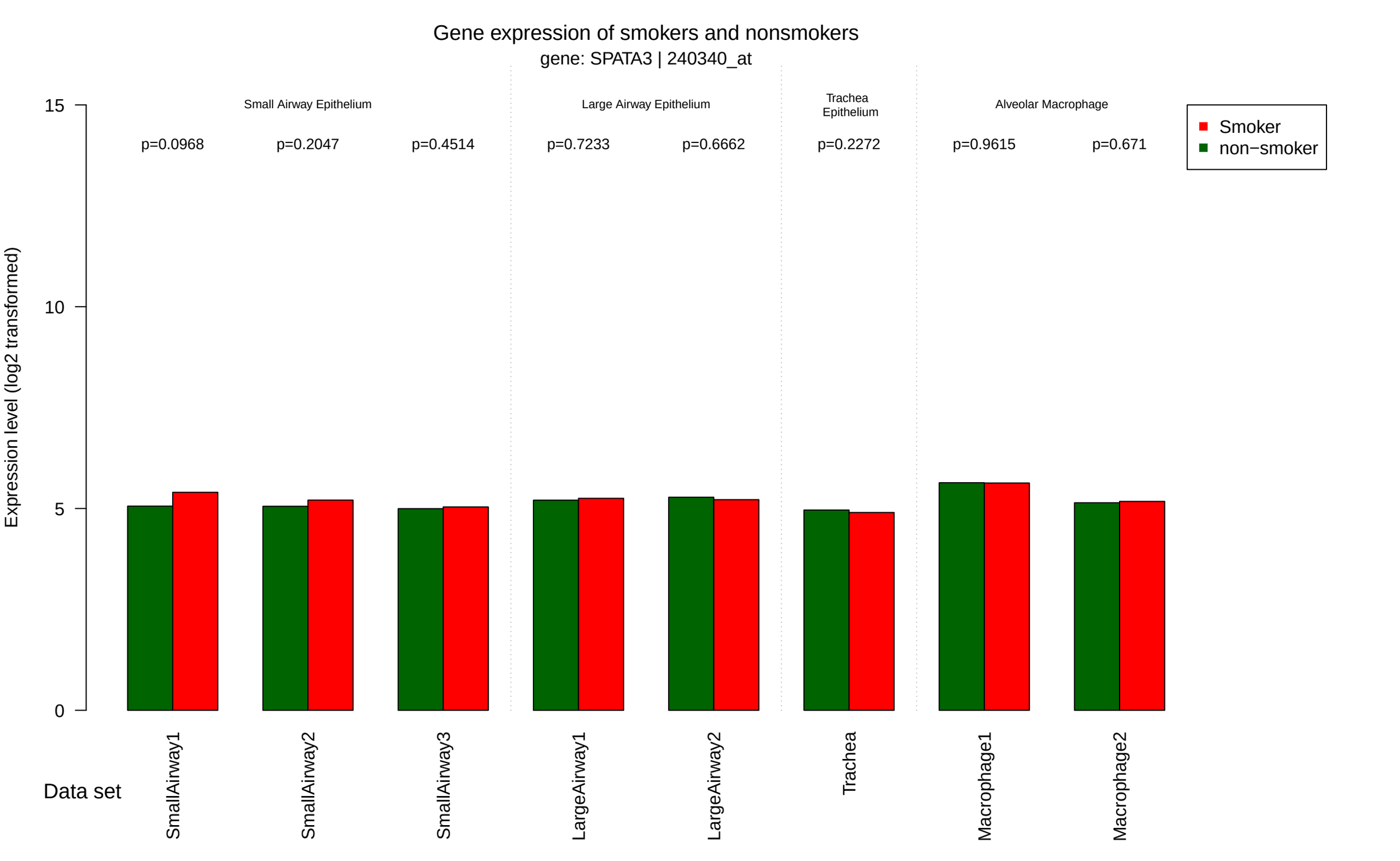
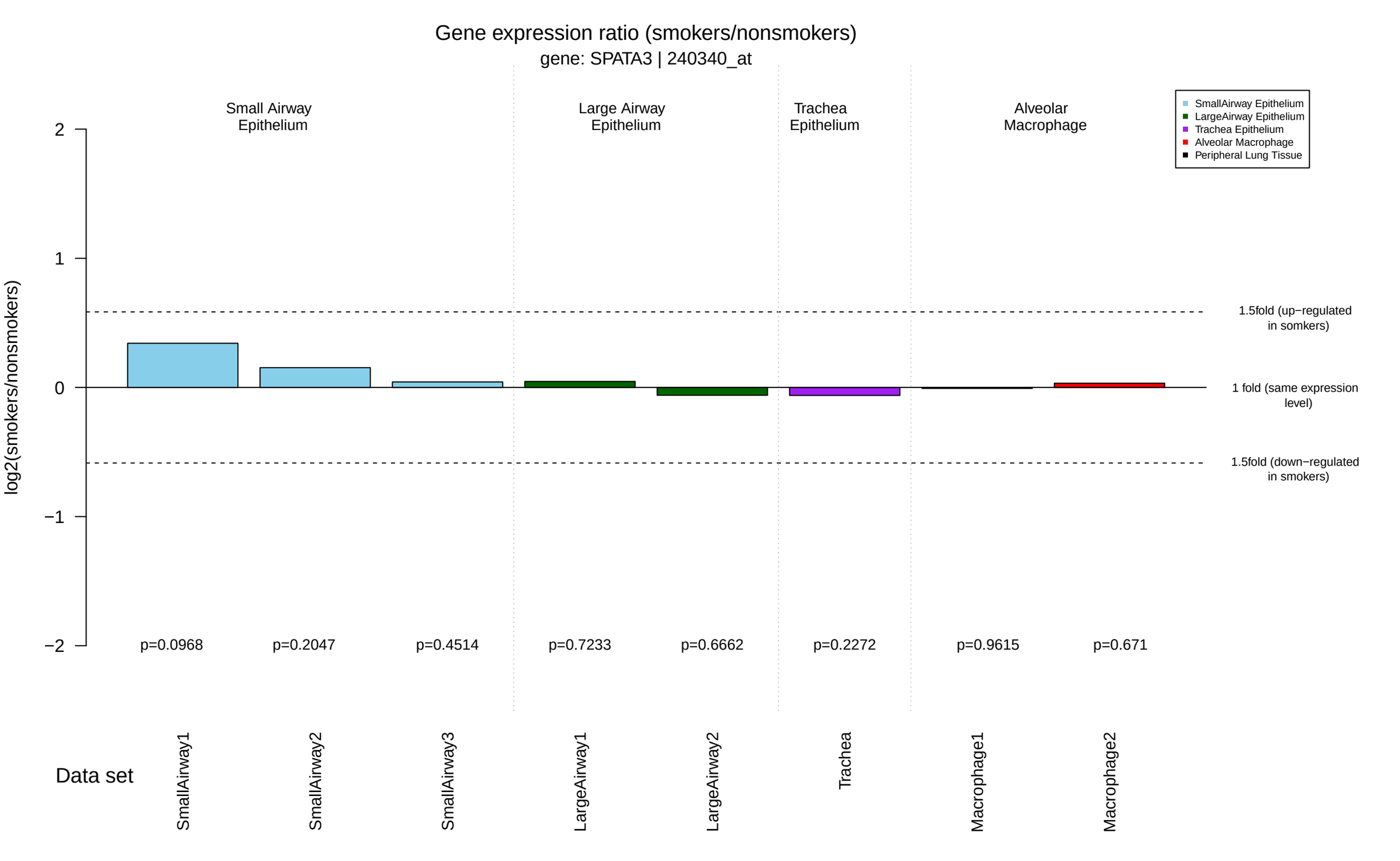
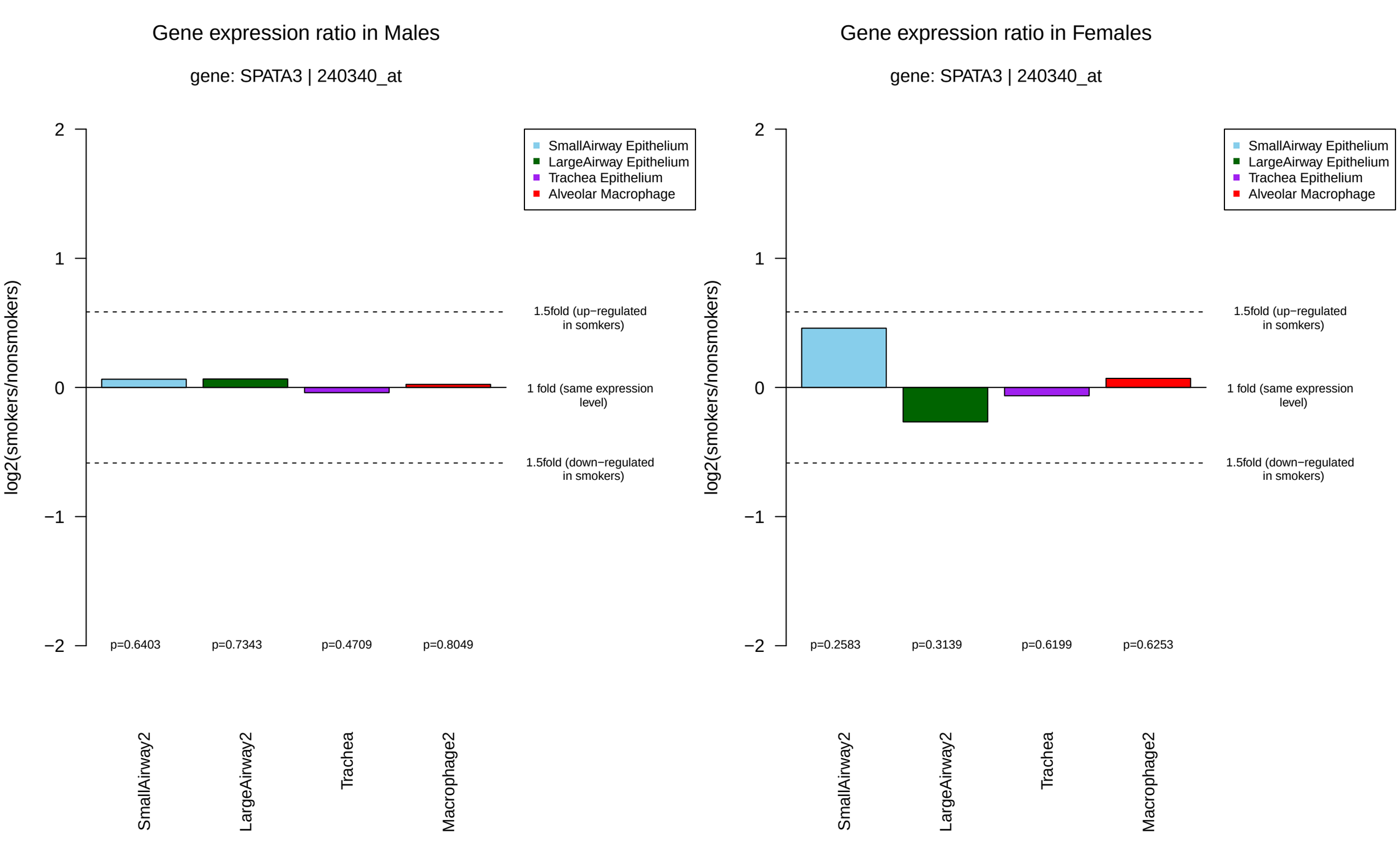
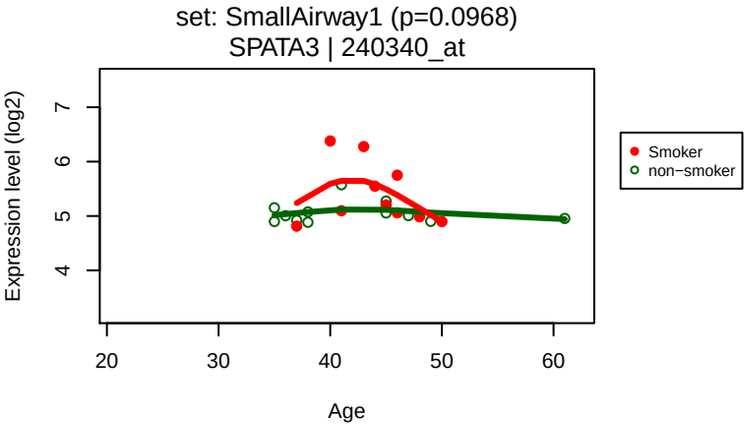
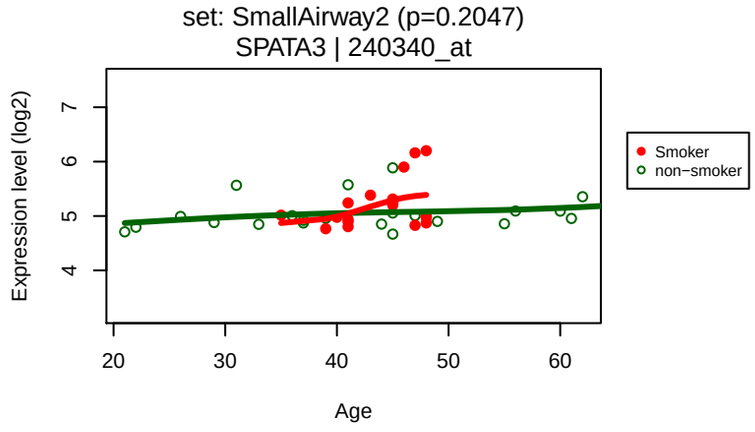
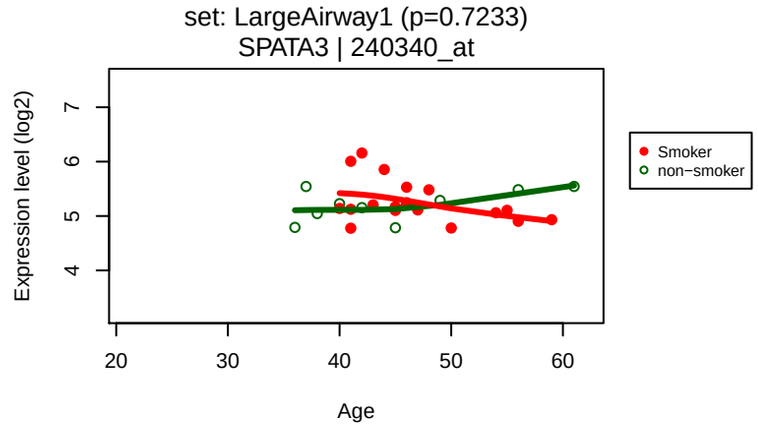

Gene expression of smokers and nonsmokers
gene: SPATA3 | 240340_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0968
p=0.2047
p=0.4514
p=0.7233
p=0.6662
p=0.2272
p=0.9615
p=0.671
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

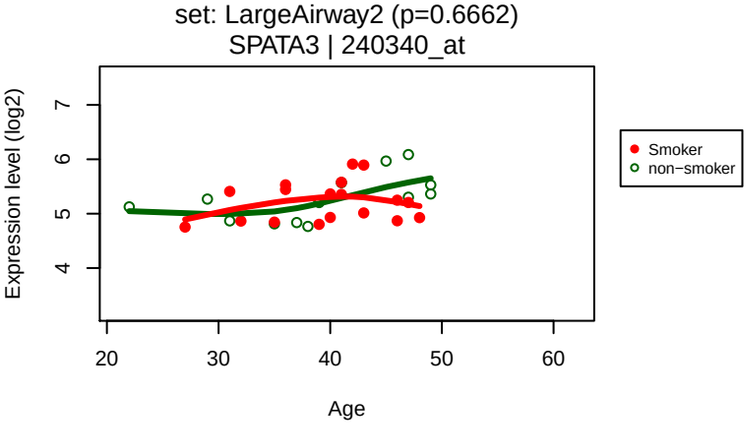

Gene expression ratio (smokers/nonsmokers)
gene: SPATA3 | 240340_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0968
p=0.2047
p=0.4514
p=0.7233
p=0.6662
p=0.2272
p=0.9615
p=0.671
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

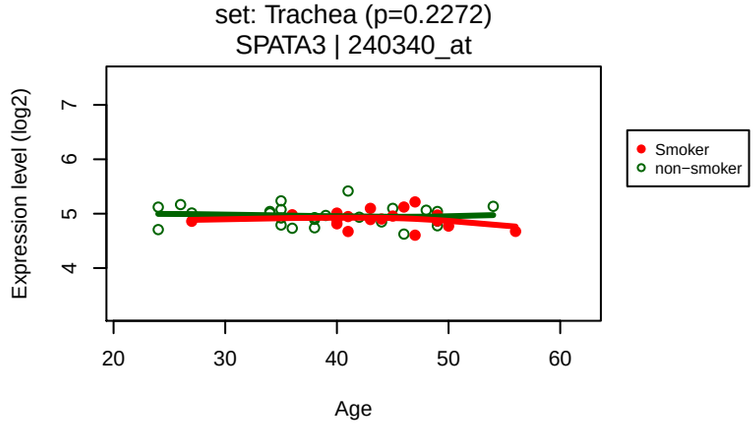

Gene expression ratio in Males
gene: SPATA3 | 240340_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6403
p=0.7343
p=0.4709
p=0.8049
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATA3 | 240340_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2583
p=0.3139
p=0.6199
p=0.6253
SmallAirway2
LargeAirway2
Trachea
Macrophage2

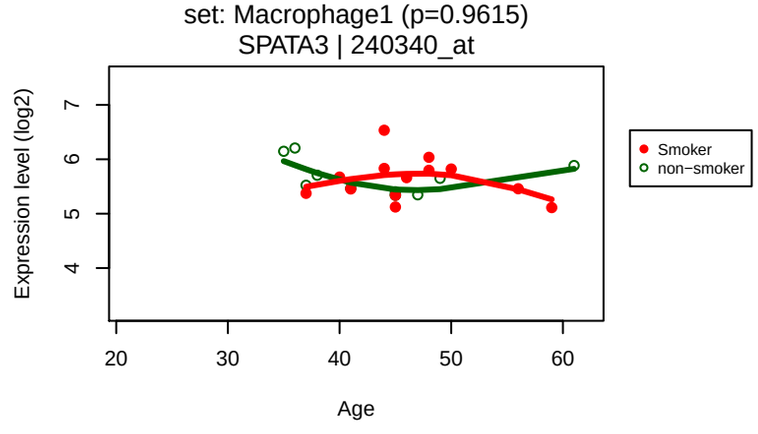

set: SmallAirway1 (p=0.0968)
SPATA3 | 240340_at
Expression level (log2)
Age
Smoker
non−smoker

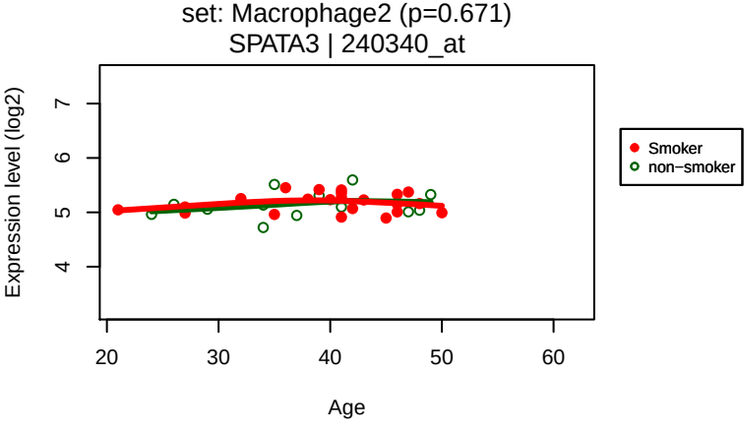

set: SmallAirway2 (p=0.2047)
SPATA3 | 240340_at
Expression level (log2)
Age
Smoker
non−smoker

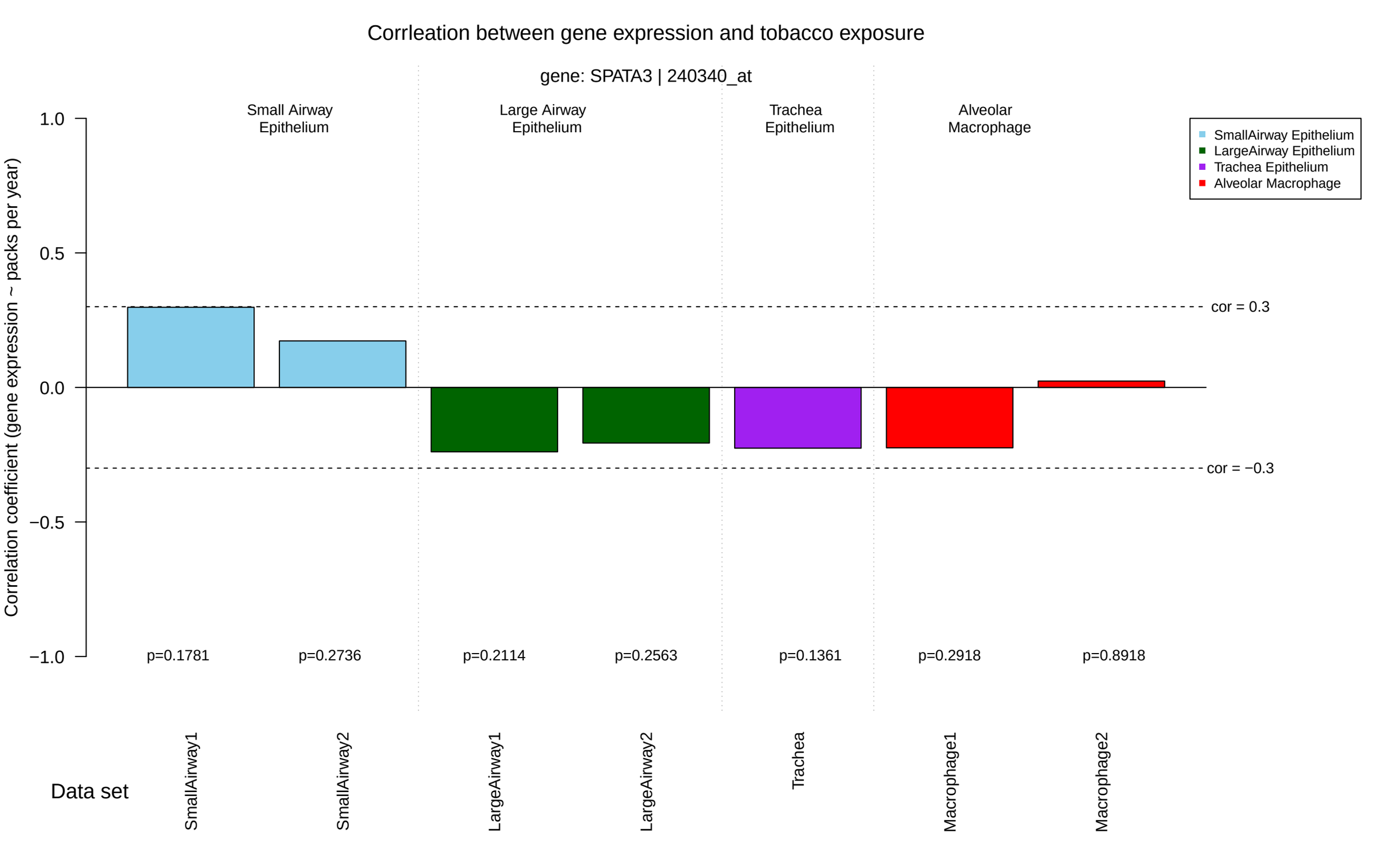

set: LargeAirway1 (p=0.7233)
SPATA3 | 240340_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6662)
SPATA3 | 240340_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2272)
SPATA3 | 240340_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9615)
SPATA3 | 240340_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.671)
SPATA3 | 240340_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA3 | 240340_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1781
p=0.2736
p=0.2114
p=0.2563
p=0.1361
p=0.2918
p=0.8918
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2